

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B.Tech. (Backlog)****University Examination May 2011****EE-208 Micro Electronics & Integrated Circuit****Date: 20/05/2011, Friday****Time: 10:00 a.m. to 1.00p.m.****Maximum Marks:70****Instructions:**

1. Attempt all questions.
2. Figure to the right indicates full marks.
3. Assume suitable data, if necessary and mention it.
4. Attempt both section separately.

SECTION- I

- Q.1 (a) Draw and Explain the block diagram of op-amp? 06**
- (b) Derive the expression for the closed loop voltage gain of inverting amplifier 04**
- (c) Define the following terms. 05**
- (1) CMRR (2) Slew Rate (3) PSRR
(4) Differential Input Resistance (5) Output Resistance

- Q.2 (a) List out the ideal characteristics of op-amp with explanation. 06**
- (b) Explain voltage to current converter using grounded load. 04**

OR

- Q.2 (a) Explain second order low pass butter worth filter with necessary circuit diagram. 06**
- (b) Explain square wave generator using op-amp. 04**
- Q.3 (a) Explain astablemultivibrator using IC-555 with suitable waveform and derive the equation of total time T. 06**
- (b) What is a voltage regulator? List four different types of voltage regulators 04**

OR

- Q.3 (a) Write a short notes: 06**
- I) Working of Schmitt trigger.
II) Summing, Scaling and averaging amplifier.
- (b) Explain integrator and differentiator using op-amp. 04**

SECTION – II

Q.4 (a) Draw and explain common-emitter transistor configuration & show its input-output characteristics. 08

(b) What are the different requirements of faithful biasing? Explain base resistor biasing methods. 07

Q.5 (a) Explain working principle of MOSFET and describe what is the difference between JFET & MOSFET. 06

(b) Define Pinch-off voltage in JFET. 04

OR

Q.5 (a) Define the following terms related to transistor 06

- 1) Cut-off point
- 2) Saturation point
- 3) Active region

(b) Write difference between JFET and bipolar transistor. 04

Q.6 (a) Describe h-parameter of a transistor and its circuit performance. 06

(b) In a common base connection, current amplification factor is 0.9. If the Emitter current is 1mA, Determine the value of base current. 04

OR

Q.6 (a) A transistor used in CE connection has the following set of h parameters 06

When the d.c. operating point is $V_{CE} = 5$ volts and $I_C = 1\text{mA}$.

$h_{ie} = 1700 \Omega$; $h_{re} = 1.3 \times 10^{-4}$; $h_{fe} = 38$; $h_{oe} = 6 \times 10^{-6} \text{mho}$.

If the a.c. load r_L seen by the transistor is 2kohm. Find (i) the input impedance (ii) Current gain and (iii) Voltage gain.

(b) For a transistor, $\beta = 45$ and voltage drop across $1\text{k}\Omega$ which is connected in The collector circuit is 1 volt. Find the base current for common emitter Connection 04